



Brings you Metabolomics and Lipidomics



"Intégration multi-omique par stratégie multiblocks et visualisation en réseaux de corrélations ou d'ontologies"

jean-charles.martin@univ-amu.fr



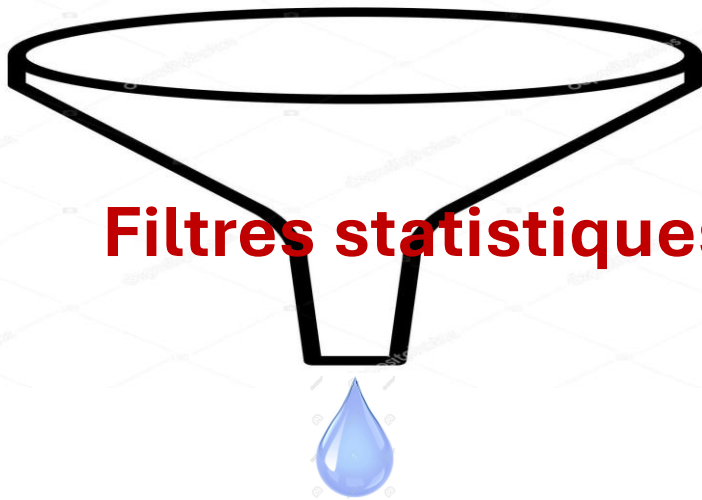
Issue#1 Metabolomics rains cats and dogs of data

1217 annotated variables

- 263 metabolites in plasma
- 136 métabolites in cortex
- 260 lipides in plasma
- 556 lipides in cortex

49 rats

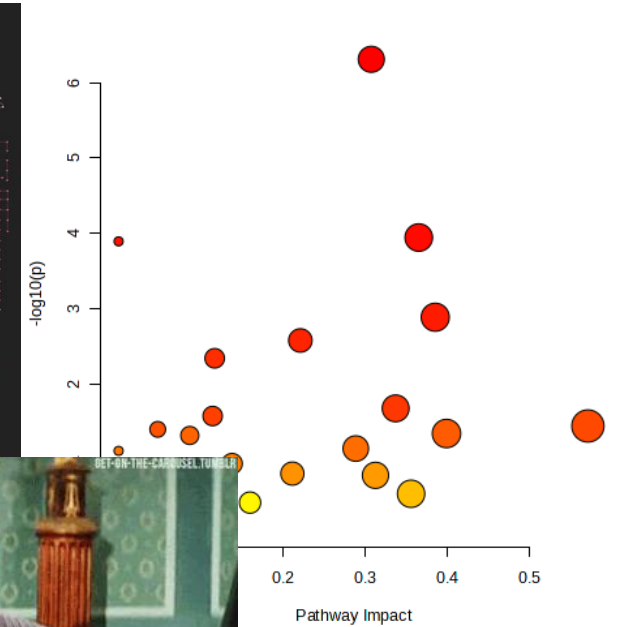
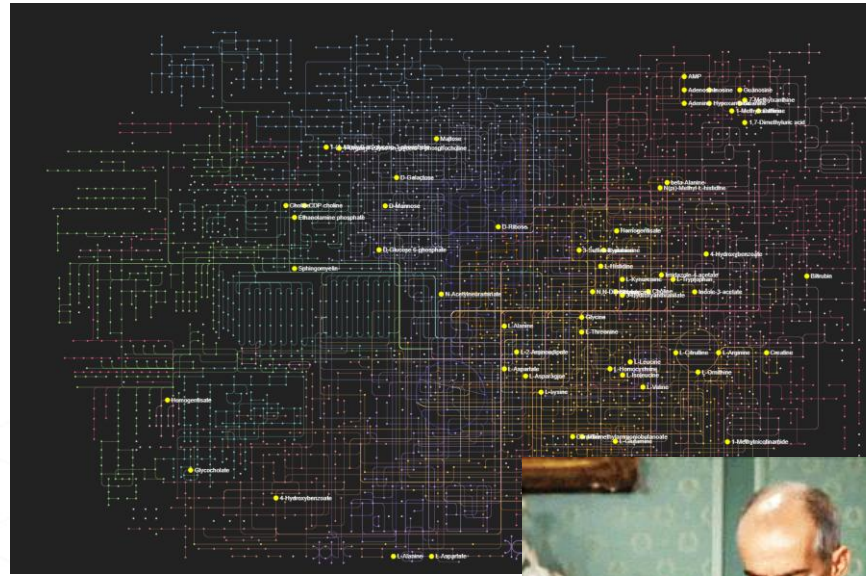
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR	AS	AT	AU	AV	AW	AX	AY	AZ	BA	BB	BC	BD	BE	BF	BG	BH	BI	BJ	BK	BL	BM	BN	BO	BP	BQ	BR	BS	BT	BU	BV	BW	BX	BY	BZ	CA	CB	CC	CD	CE	CF	CG	CH	CI	CJ	CK	CL	CM	CN	CO	CP	CQ	CR	CS	CT	CU	CV	CW	CX	CY	CZ	DA	DB	DC	DD	DE	DF	DG	DH	DI	DJ	DK	DL	DM	DN	DO	DP	DQ	DR	DS	DT	DU	DV	DW	DX	DY	DZ	EA	EB	EC	ED	EE	EF	EG	EH	EI	EJ	EK	EL	EM	EN	EO	EP	EQ	ER	ES	ET	EU	EV	EW	EX	EY	EZ	FA	FB	FC	FD	FE	FF	FG	FH	FI	FJ	FK	FL	FM	FN	FO	FP	FQ	FR	FS	FT	FU	FV	FW	FX	FY	FZ	GA	GB	GC	GD	GE	GF	GG	GH	GI	GJ	GK	GL	GM	GN	GO	GP	GQ	GR	GS	GT	GU	GV	GW	GX	GY	GZ	HA	HB	HC	HD	HE	HF	HG	HH	HI	HJ	HK	HL	HM	HN	HO	HP	HQ	HR	HS	HT	HU	HV	HW	HX	HY	HZ	IA	IB	IC	ID	IE	IF	IG	IH	II	IJ	IK	IL	IM	IN	IO	IP	IQ	IR	IS	IT	IU	IV	IW	IX	IY	IZ	JA	JB	JC	JD	JE	JF	JG	JH	JI	IJ	JK	JL	JM	JN	JO	JP	JQ	JR	JS	JT	JU	JV	JW	JX	JY	JZ	KA	KB	KC	KD	KE	KF	KG	KH	KI	KJ	KK	KL	KM	KN	KO	KP	KQ	KR	KS	KT	KU	KV	KW	KX	KY	KZ	LA	LB	LC	LD	LE	LF	LG	LH	LI	LJ	LK	LM	LN	LO	LP	LQ	LR	LS	LT	LU	LV	LW	LX	LY	LZ	MA	MB	MC	MD	ME	MF	MG	MH	MI	MJ	MK	ML	MM	MN	MO	MP	MQ	MR	MS	MT	MU	MV	MW	MX	MY	MZ	NA	NB	NC	ND	NE	NF	NG	NH	NI	NJ	NK	NL	NM	NN	NO	NP	NQ	NR	NS	NT	NU	NV	NW	NX	NY	NZ	OA	OB	OC	OD	OE	OF	OG	OH	OI	OJ	OK	OL	OM	ON	OO	OP	OQ	OR	OS	OT	OU	OV	OW	OX	OY	OZ	PA	PB	PC	PD	PE	PF	PG	PH	PI	PJ	PK	PL	PM	PN	PO	PP	PQ	PR	PS	PT	PU	PV	PW	PX	PY	PZ	QA	QB	QC	QD	QE	QF	QG	QH	QI	QJ	QK	QL	QM	QN	QO	QP	QQ	QR	QS	QT	QU	QV	QW	QX	QY	QZ	RA	RB	RC	RD	RE	RF	RG	RH	RI	RJ	RK	RL	RM	RN	RO	RP	RQ	RR	RS	RT	RU	RV	RW	RX	RY	RZ	SA	SB	SC	SD	SE	SF	SG	SH	SI	SJ	SK	SL	SM	SN	SO	SP	SQ	SR	SS	ST	SU	SV	SW	SX	SY	SZ	TA	TB	TC	TD	TE	TF	TG	TH	TI	TJ	TK	TL	TM	TN	TO	TP	TQ	TR	TS	TU	TV	TW	TX	TY	TZ	UA	UB	UC	UD	UE	UF	UG	UH	UI	UJ	UK	UL	UM	UN	UO	UP	UQ	UR	US	UT	UU	UV	UW	UX	UY	UZ	VA	VB	VC	VD	VE	VF	VG	VH	VI	VJ	VK	VL	VM	VN	VO	VP	VQ	VR	VS	VT	VU	VV	VW	VX	VY	VZ	WA	WB	WC	WD	WE	WF	WG	WH	WI	WJ	WK	WL	WM	WN	WO	WP	WQ	WR	WS	WT	WU	WV	WW	WX	WY	WZ	XA	XB	XC	XD	XE	XF	YG	YH	YI	YJ	YK	YL	YM	YN	YO	YP	YQ	YR	YS	YT	YU	YV	YW	YX	YY	YZ	ZA	ZB	ZC	ZD	ZE	ZF	ZG	ZH	ZI	ZJ	ZK	ZL	ZM	ZN	ZO	ZP	ZQ	ZR	ZS	ZT	ZU	ZV	ZW	ZX	ZY	ZZ	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR	AS	AT	AU	AV	AW	AX	AY	AZ	BA	BB	BC	BD	BE	BF	BG	BH	BI	BJ	BK	BL	BM	BN	BO	BP	BQ	BR	BS	BT	BU	BV	BW	BX	BY	BZ	CA	CB	CC	CD	CE	CF	CG	CH	CI	CJ	CK	CL	CM	CN	CO	CP	CQ	CR	CS	CT	CU	CV	CW	CX	CY	CZ	DA	DB	DC	DD	DE	DF	DG	DH	DI	DJ	DK	DL	DM	DN	DO	DP	DQ	DR	DS	DT	DU	DV	DW	DX	DY	DZ	EA	EB	EC	ED	EE	EF	EG	EH	EI	EJ	EK	EL	EM	EN	EO	EP	EQ	ER	ES	ET	EU	EV	EW	EX	EY	EZ	FA	FB	FC	FD	FE	FF	FG	FH	FI	FJ	FK	FL	FM	FN	FO	FP	FQ	FR	FS	FT	FU	FV	FW	FX	FY	FZ	GA	GB	GC	GD	GE	GF	GG	GH	GI	GJ	GK	GL	GM	GN	GO	GP	GQ	GR	GS	GT	GU	GV	GW	GX	GY	GZ	HA	HB	HC	HD	HE	HF	HG	HH	HI	HJ	HK	HL	HM	HN	HO	HP	HQ	HR	HS	HT	HU	HV	HW	HX	HY	HZ	IA	IB	IC	ID	IE	IF	IG	IH	II	IJ	IK	IL	IM	IN	IO	IP	IQ	IR	IS	IT	IU	IV	IW	IX	IY	IZ	JA	JB	JC	JD	JE	JF	JG	JH	JI	IJ	JK	JL	JM	JN	JO	JP	JQ	JR	JS	JT	JU	JV	JW	JX	JY	JZ	KA	KB	KC	KD	KE	KF	KG	KH	KI	KJ	KK	KL	KM	KN	KO	KP	KQ	KR	KS	KT	KU	KV	KW	KX	KY	KZ	LA	LB	LC	LD	LE	LF	LG	LH	LI	LJ	LK	LM	LN	LO	LP	LQ	LR	LS	LT	LU	LV	LW	LX	LY	LZ	MA	MB	MC	MD	ME	MF	MG	MH	MI	MJ	MK	ML	MM	MN	MO	MP	MQ	MR	MS	MT	MU	MV	MW	MX	MY	MZ	NA	NB	NC	ND	NE	NF	NG	NH	NI	NJ	NK	NL	NM	NN	NO	NP	NQ	NR	NS	NT	NU	NV	NW	NX	NY	NZ	OA	OB	OC	OD	OE	OF	OG	OH	OI	OJ	OK	OL	OM	ON	OO	OP	OQ	OR	OS	OT	OU	OV	OW	OX	OY	OZ	PA	PB	PC	PD	PE	PF	PG	PH	PI	PJ	PK	PL	PM	PN	PO	PP	PQ	PR	PS	PT	PU	PV	PW	PX	PY	PZ	QA	QB	QC	QD	QE	QF	QG	QH	QI	QJ	QK	QL	QM	QN	QO	QP	QQ	QR	QS	QT	QU	QV	QW	QX	QY	QZ	RA	RB	RC	RD	RE	RF	RG	RH	RI	RJ	RK	RL	RM	RN	RO	RP	RQ	RR	RS	RT	RU	RV	RW	RX	RY	RZ	SA	SB	SC	SD	SE	SF	SG	SH	SI	SJ	SK	SL	SM	SN	SO	SP	SQ	SR	SS	ST	SU	SV	SW	SX	SY	SZ	TA	TB	TC	TD	TE	TF	TG	TH	TI	TJ	TK	TL	TM	TN	TO	TP	TQ	TR	TS	TU	TV	TW	TX	TY	TZ	UA	UB	UC	UD	UE	UF	UG	UH	UI	UJ	UK	UL	UM	UN	UO	UP	UQ	UR	US	UT	UU	UV	UW	UX	UY	UZ	VA	VB	VC	VD	VE	VF	VG	VH	VI	VJ	VK	VL	VM	VN	VO	VP	VQ	VR	VS	VT	VU	VV	VW	VX	VY	VZ	WA	WB	WC	WD	WE	WF	WG	WH	WI	WJ	WK	WL	WM	WN	WO	WP	WQ	WR	WS	WT	WU	WV	WW	WX	WY	WZ	XA	XB	XC	XD	XE	XF	YG	YH	YI	YJ	YK	YL	YM	YN	YO	YP	YQ	YR	YS	YT	YU	YV	YW	YX	YY	YZ	ZA	ZB	ZC	ZD	ZE	ZF	ZG	ZH	ZI	ZJ	ZK	ZL	ZM	ZN	ZO	ZP	ZQ	ZR	ZS	ZT	ZU	ZV	ZW	ZX	ZY	ZZ	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR	AS	AT	AU	AV	AW	AX	AY	AZ	BA	BB	BC	BD	BE	BF	BG	BH	BI	BJ	BK	BL	BM	BN	BO	BP	BQ	BR	BS	BT	BU	BV	BW	BX	BY	BZ	CA	CB	CC	CD	CE	CF	CG	CH	CI	CJ	CK	CL	CM	CN	CO	CP	CQ	CR	CS	CT	CU	CV	CW	CX	CY	CZ	DA	DB	DC	DD	DE	DF	DG	DH	DI	DJ	DK	DL	DM	DN	DO	DP	DQ	DR	DS	DT	DU	DV	DW	DX	DY	DZ	EA	EB	EC	ED	EE	EF	EG	EH	EI	EJ	EK	EL	EM	EN	EO	EP	EQ	ER	ES	ET	EU	EV	EW	EX	EY	EZ	FA	FB	FC	FD	FE	FF	FG	FH	FI	FJ	FK	FL	FM	FN	FO	FP	FQ	FR	FS	FT	FU	FV	FW	FX	FY	FZ	GA	GB	GC	GD	GE	GF	GG	GH	GI	GJ	GK	GL	GM	GN	GO	GP	GQ	GR	GS	GT	GU	GV	GW	GX	GY	GZ	HA	HB	HC	HD	HE	HF	HG	HH	HI	HJ	HK	HL	HM	HN	HO	HP	HQ	HR	HS	HT	HU	HV	HW	HX	HY	HZ	IA	IB	IC	ID	IE	IF	IG	IH	II	IJ	IK	IL	IM	IN	IO	IP	IQ	IR	IS	IT	IU	IV	IW	IX	IY	IZ	JA	JB	JC	JD	JE	JF	JG	JH	JI	IJ	JK	JL	JM	JN	JO	JP	JQ	JR	JS	JT	JU	JV	JW	JX	JY	JZ	KA	KB	KC	KD	KE	KF	KG	KH	KI	KJ	KK	KL	KM	KN	KO	KP	KQ	KR	KS	KT	KU	KV	KW	KX	KY	KZ	LA	LB	LC	LD	LE	LF	LG	LH	LI	LJ	LK	LM	LN	LO	LP	LQ	LR	LS	LT	LU	LV	LW	LX	LY	LZ	MA	MB	MC	MD	ME	MF	MG	MH	MI	MJ	MK	ML	MM	MN	MO	MP	MQ	MR	MS	MT	MU	MV	MW	MX	MY	MZ	NA	NB	NC	ND	NE	NF	NG	NH	NI	NJ	NK	NL	NM	NN	NO	NP	NQ	NR	NS	NT	NU	NV	NW	NX	NY	NZ	OA	OB	OC	OD	OE	OF	OG	OH	OI	OJ	OK	OL	OM	ON	OO	OP	OQ	OR	OS	OT	OU	OV	OW	OX	OY	OZ	PA	PB	PC	PD	PE	PF	PG	PH	PI	PJ	PK	PL	PM	PN	PO	PP	PQ	PR	PS	PT	PU	PV	PW	PX	PY	PZ	QA	QB	QC	QD	QE	QF	QG	QH	QI	QJ	QK	QL	QM	QN	QO	QP	QQ	QR	QS	QT	QU	QV	QW	QX	QY	QZ	RA	RB	RC	RD	RE	RF	RG	RH	RI	RJ	RK	RL	RM	RN	RO	RP	RQ	RR	RS	RT	RU	RV	RW	RX	RY	RZ	SA	SB	SC	SD	SE	SF	SG	SH	SI	SJ	SK	SL	SM	SN	SO	SP	SQ	SR	SS	ST	SU	SV	SW	SX	SY	SZ	TA	TB	TC	TD	TE	TF	TG	TH	TI	TJ	TK	TL	TM	TN	TO	TP	TQ	TR	TS	TU	TV	TW	TX	TY	TZ	UA	UB	UC	UD	UE	UF	UG	UH	UI	UJ	UK	UL	UM	UN	UO	UP	UQ	UR	US	UT	UU	UV	UW	UX	UY	UZ	VA	VB	VC	VD	VE	VF	VG	VH	VI	VJ	VK	VL	VM	VN	VO	VP	VQ	VR	VS	VT	VU	VV	VW	VX	VY	VZ	WA	WB	WC	WD	WE	WF	WG	WH	WI	WJ	WK	WL	WM	WN	WO	WP	WQ	WR	WS	WT	WU	WV	WW	WX	WY	WZ	XA	XB	XC	XD	XE	XF	YG	YH	YI	YJ	YK	YL	YM	YN	YO	YP	YQ	YR	YS	YT	YU	YV	YW	YX	YY	YZ	ZA	ZB	ZC	ZD	ZE	ZF	ZG	ZH	ZI	ZJ	ZK	ZL	ZM	ZN	ZO	ZP	ZQ	ZR	ZS	ZT	ZU	ZV	ZW	ZX	ZY	ZZ	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR	AS	AT	AU	AV	AW	AX	AY	AZ	BA	BB	BC	BD	BE	BF	BG	BH	BI	BJ	BK	BL	BM	BN	BO	BP	BQ
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Filtres statistiques

« statistically » relevant metabolites

C03197
C16356
C08317
C02494
C02918
C16359
C16358
C21298
C01987
C00550
C02323
C02735
C02504
C01198
C16360
C12115
C00632
C00606
C21167
C00156
C01181
C16353
C00147
C00212
C00020
C00956
C00099
C00719
C00486
C02862
C07481
C00114
C00307
C00327
C00300
C00791
C03299
C05422
C04483
C00124
C06772
C01026
C00077



Pathway Name						DR	Impact	Details
Caffeine metabolism	6/10	4.9149E-7	6.3085	3.9319E-5	3.9319E-5	0.30769	KEGG SMP	
Arginine biosynthesis	5/14	1.1316E-4	3.9463	0.0089394	0.0033904	0.36548	KEGG	
Valine, leucine and isoleucine biosynthesis	4/8	1.2714E-4	3.8957	0.0099169	0.0033904	0.0	KEGG SMP	
Glycine, serine and threonine metabolism	6/33	0.0012869	2.8904	0.099095	0.025739	0.38556	KEGG SMP	
Histidine metabolism	4/16	0.0026152	2.5825	0.19875	0.041843	0.22131	KEGG SMP	
Purine metabolism	8/70	0.0045068	2.3461	0.33801	0.06009	0.11699	KEGG SMP	

<https://doi.org/10.1038/s42255-025-01283-0>

Metabolites are not genes – avoiding the misuse of pathway analysis in metabolomics

Kil Sun Lee, Xiaoyang Su & Tao Huan

Pathway analysis, originally developed for gene expression data, has been adapted for metabolomics. However, owing to the unique characteristics and constraints of metabolites compared with genes, pathway analysis in metabolomics often yields misleading or nonsensical results.

Lee KS, Su X, Huan T. Metabolites are not genes — avoiding the misuse of pathway analysis in metabolomics. Nature Metabolism. 2025 2025/04/10.

issues are well documented and broadly acknowledged elsewhere, they are often overlooked by bioinformaticians with a limited understanding of the distinct biological and chemical characteristics of genes and metabolites. They are also underestimated by experimental scientists, who frequently rely on the tools developed by bioinformaticians without fully grasping their underlying mechanisms.

When a pathway is upregulated, all the genes involved in that pathway produce more copies. However, in linear biosynthetic pathways, metabolite levels show lower coherence during regulation

Increased pathway activity may be masked by high metabolite export rates

Metabolites are frequently transported between organs and tissues, so the metabolic concentration does not reflect the activity of a particular gene

single metabolite can arise from multiple metabolic activities and serve diverse functions. Unlike genes, metabolites do not adhere to the generally valid 'one gene, one function' principle

MS-based metabolomics has incomplete metabolome coverage

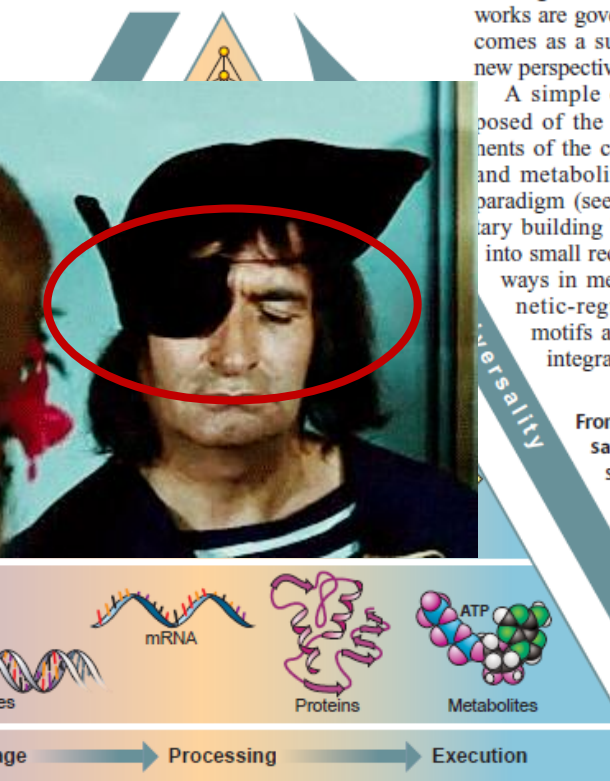
Metabolites can arise from exogenous sources (diet, micro-organisms...)

Life's Complexity Pyramid

Zoltán N. Oltvai and Albert-László Barabási

Cells and microorganisms have an impressive capacity for adjusting their intracellular machinery in response to changes in their environment, food availability, and developmental state. Add to this an amazing ability to correct internal errors—battling the effects of such mistakes as mutations or misfolded proteins—and we arrive at a major issue of contemporary cell biology: our need to comprehend the staggering complexity, versatility, and robustness of living systems. Although molecular biology offers many spectacular successes, it is clear that the detailed inventory of genes, proteins, and metabolites is not sufficient to understand the cell's complexity (1). As demonstrated by two papers in this issue—Lee *et al.* (2) on page 799 and Milo *et al.* (3) on page 824—viewing the cell as a network of genes and proteins offers a viable strategy for addressing the complexity of living systems.

According to the basic dogma of molecular biology, DNA is the ultimate depository of biological complexity. Indeed, it is generally accepted that information storage, information processing, and the execution of various cellular programs reside in distinct levels of organization: the cell's genome, transcriptome, proteome, and



within large networks (6, 7). There is clear evidence for the existence of such cellular networks: For example, the proteome organizes itself into a protein interaction network and metabolites are interconverted through an intricate metabolic web (7). The finding that the structures of these networks are governed by the same principles comes as a surprise, however, offering a new perspective on cellular organization.

A simple complexity pyramid composed of the various molecular components of the cell—genes, RNAs, proteins, and metabolites—summarizes this new paradigm (see the figure). These elementary building blocks organize themselves into small recurrent patterns, called pathways in metabolism and motifs in genetic-regulatory networks. In turn, motifs and pathways are seamlessly integrated to form functional mod-

From the particular to the universal. The bottom of the pyramid shows the traditional representation of the cell's functional organization: genome, transcriptome, proteome, and metabolome (level 1). There is remarkable integration of the various layers both at the regulatory and the structural level. Insights into the logic of cellular organization can be achieved when we view

the cell as a complex network in which the components are connected by functional links. At the lowest level, these components form genetic-regulatory motifs or metabolic pathways (level 2), which in turn are the building blocks of functional modules (level 3). These modules are nested, generating a scale-free hierarchical architecture (level 4). Although the individual components are unique to a given organism, the topologic properties of cellular networks share surprising similarities with those of natural and social networks. This suggests that universal organizing principles apply to all networks, from the cell to the World Wide Web.

metabolome. However, the distinctness of these organizational levels has recently come under fire. For example, although long-term information is stored almost exclusively in the genome, the proteome is crucial for short-term information storage (4, 5) and transcription factor-controlled information retrieval is strongly influenced by the state of the metabolome. This integration of different organizational levels increasingly forces us to view cellular functions as distributed among groups of heterogeneous components that all interact

Z. N. Oltvai is in the Department of Pathology, Northwestern University, Chicago, IL 60611, USA. E-mail: zno008@nwu.edu A.-L. Barabási is in the Department of Physics, University of Notre Dame, Notre Dame, IN 46556, USA. E-mail: alb@nd.edu

CREDIT: KATHARINE SUTUFF/SCIENCE

Issue#2 Not significant, really?

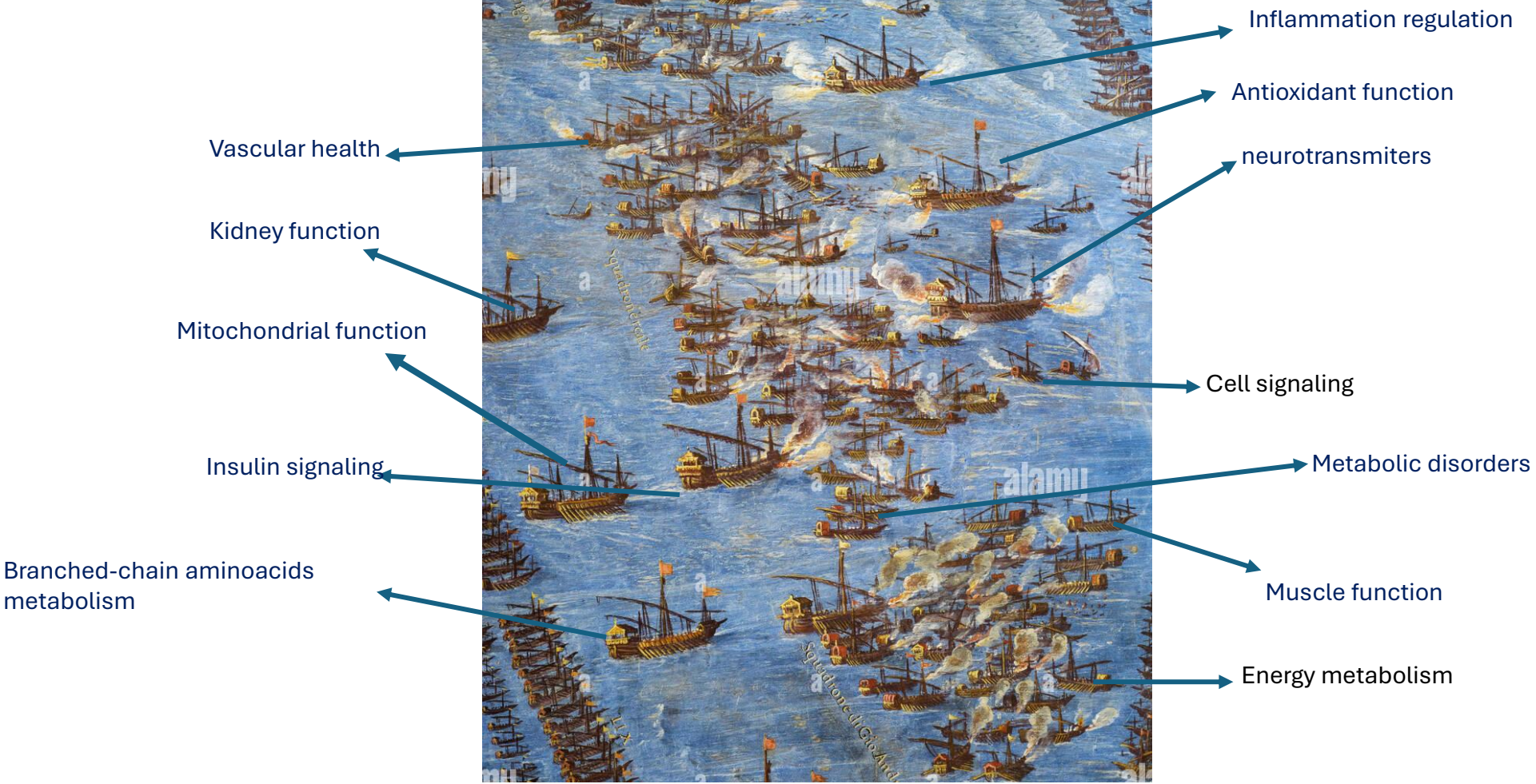


$\Rightarrow$ One rower is not enough to move the boat forward ($P_{\text{value one rower}} > 0.05$)

$\Rightarrow$ The whole team is necessary and significant ($P_{\text{value}} < 0.05$)

$\Rightarrow$ Better to consider the rowers as a team, not as separated individuals (each $P > 0.05$ gives overall $P < 0.05$)

The whole fleet is the metabolome!



Niveau moléculaire (information biochimique)

Niveau fonctionnel (information processus biologiques)

Niveau systèmes (Information sur la réponse phénotypique)

Tryptophan metabolism
Gut-liver indole metabolism
Farnesyl to CoQ10 metabolism
Amine Oxidase reactions

Phosphate co-transporter
Sodium-coupled phosphate cotransporters

Thiamine metabolism
Creatine pathway
SLC-mediated transmembrane transport
Ammonia recycling

Methylhistidine metabolism
Betaine metabolism
Methionine metabolism

SLC-mediated transmembrane transport
Purine metabolism
Nucleotide salvage
Transport of small molecules

Vitamin B6 metabolism
Steroid hormone biosynthesis
Histidine metabolism
Steroidogenesis

2 (2-AMINOETHYL)PHOSPHONATE
2 3-INDOLEPROPIONIC Acid
3-(2,4-Dihydroxyphenyl)propionic acid
3-(3-HYDROXYPHENYL)PROPIONIC acid
3,4-DIHYDROXYHYDROCINNAMIC Acid
3-AMINO-4-HYDROXYBENZOATE

Phosphoric acid
TRIGONELLINE
Mevalonolactone

OROTATE
PANTOTHENATE
PHOSPHOCREATINE
Phosphoric acid
PYRIDOXINE
RIBOFLAVIN

GLYCINE
Histamine
L-ORNITHINE
HYPOXANTHINE
INOSINE
Phosphoric acid
CYTIDINE
1-METHYLADENOSINE
3-PHENYLLACTIC Acid
FUROSE
Histamine
INDOLE 3-CARBOXALDEHYDE
LYSINE
MANNOSE
PYRIDOXAL

**Microbiota
metabolism**

kidney function

MetaboSmart™
Metabolomics Made Smart

epigenetic modification

hemostasis

immune function

Microbiota metabolic systems

Organs and tissues systems

metabolic control systems

Genomics systems

Cardiovascular systems

Cell defense systems

HIERARCHICAL MULTIBLOCK PLS AND PC MODELS FOR EASIER MODEL INTERPRETATION AND AS AN ALTERNATIVE TO VARIABLE SELECTION

SVANTE WOLD*

Research Group for Chemometrics, Umeå University, S-901 87 Umeå, Sweden

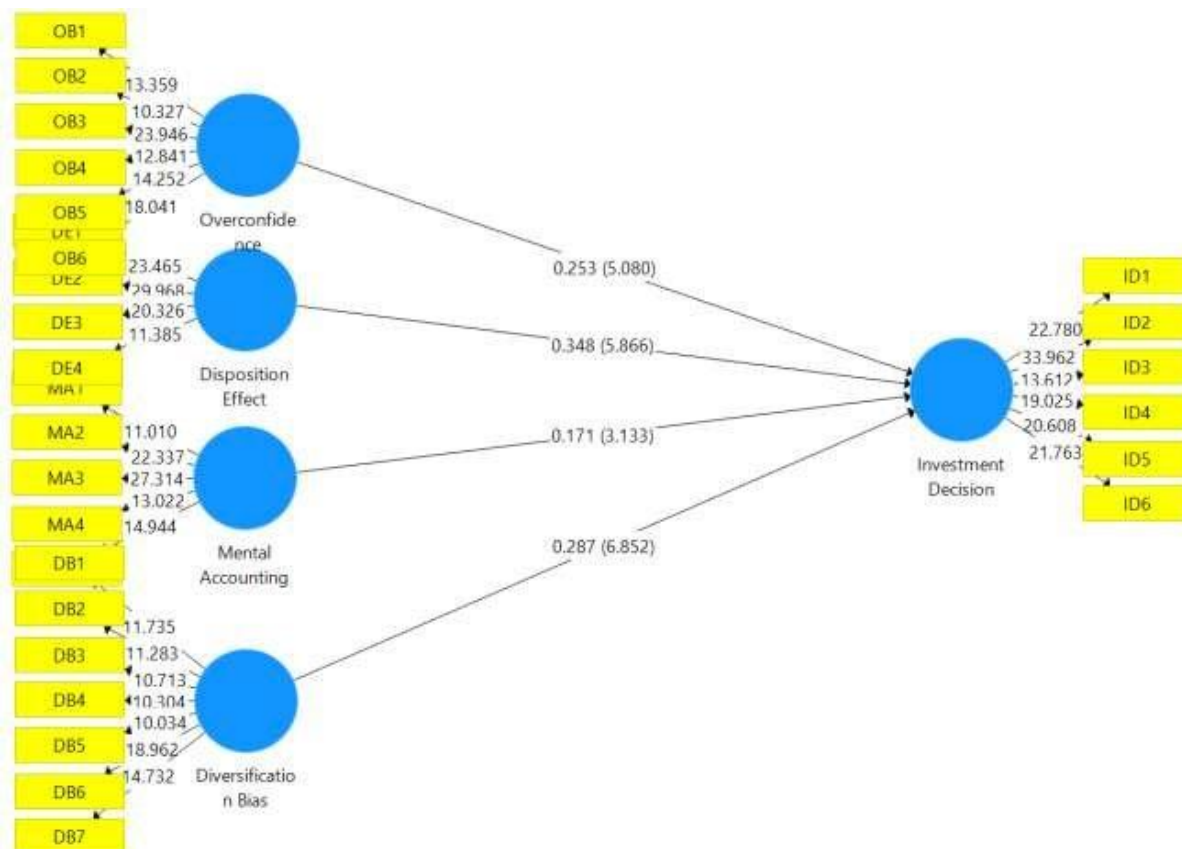
NOUNA KETTANEH

MDS Inc., 371 Highland Avenue, Winchester, MA 10890, U.S.A.

AND

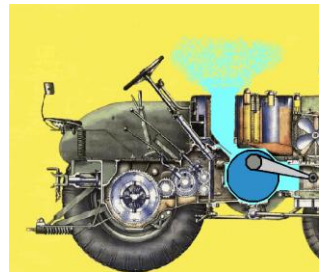
KJELL TJESSEM

Statoil, N-5154 Mongstad, Norway



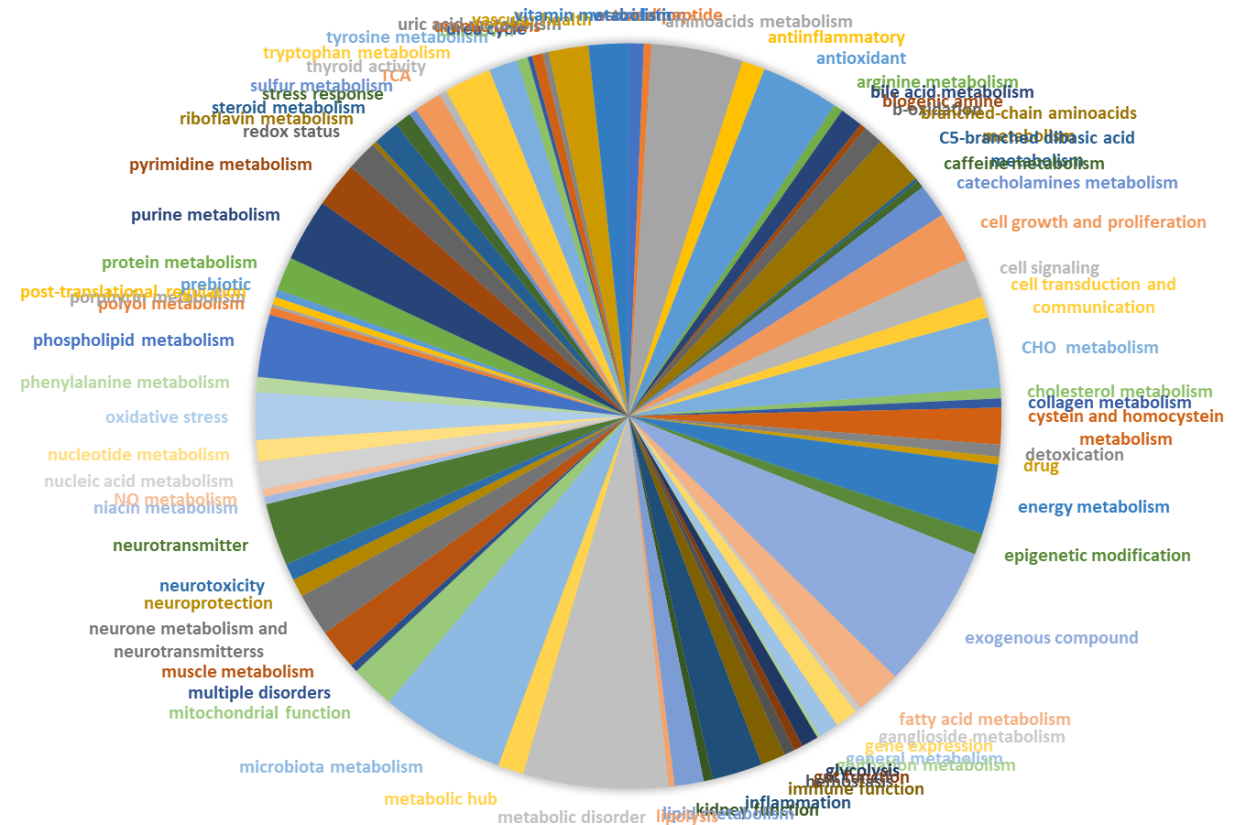
Primary metabolism: the basis of c where everything starts and ends up

The bottleneck to focus o



Primary metabolism data

- 1300 metabolites
- 100 pathways & biological functions



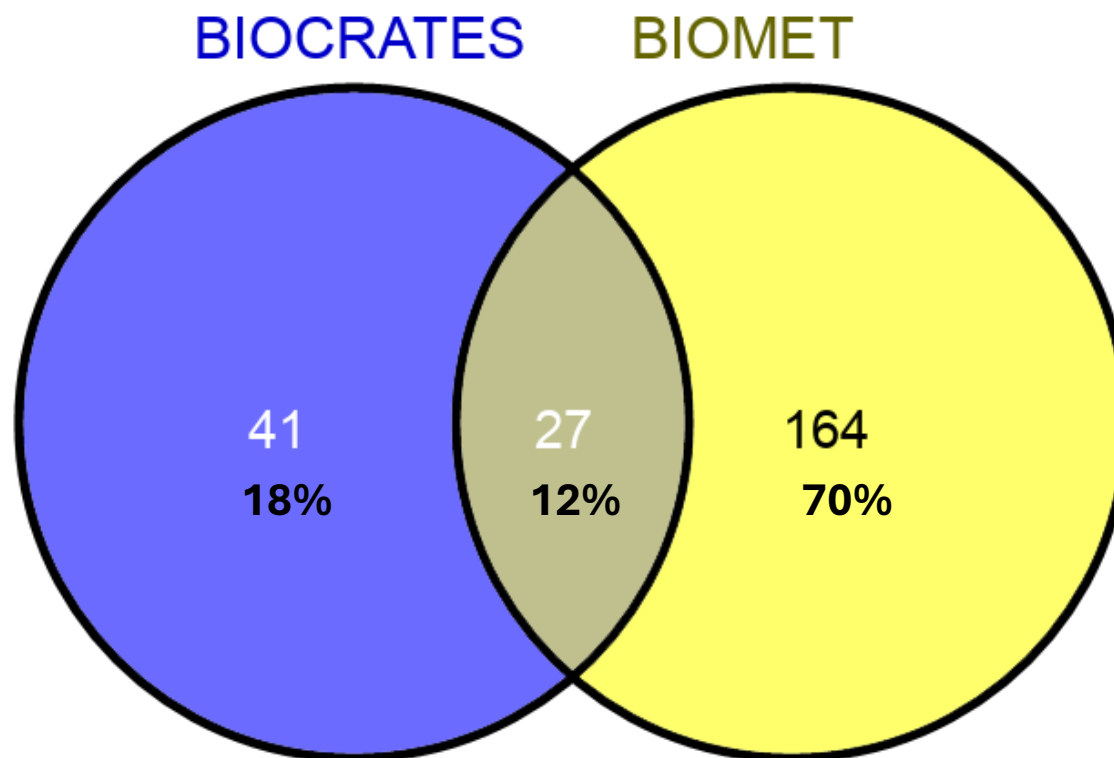
Glycan Biosynthesis and Metabolism

Biodegradation of Xenobiotics

Metabolism

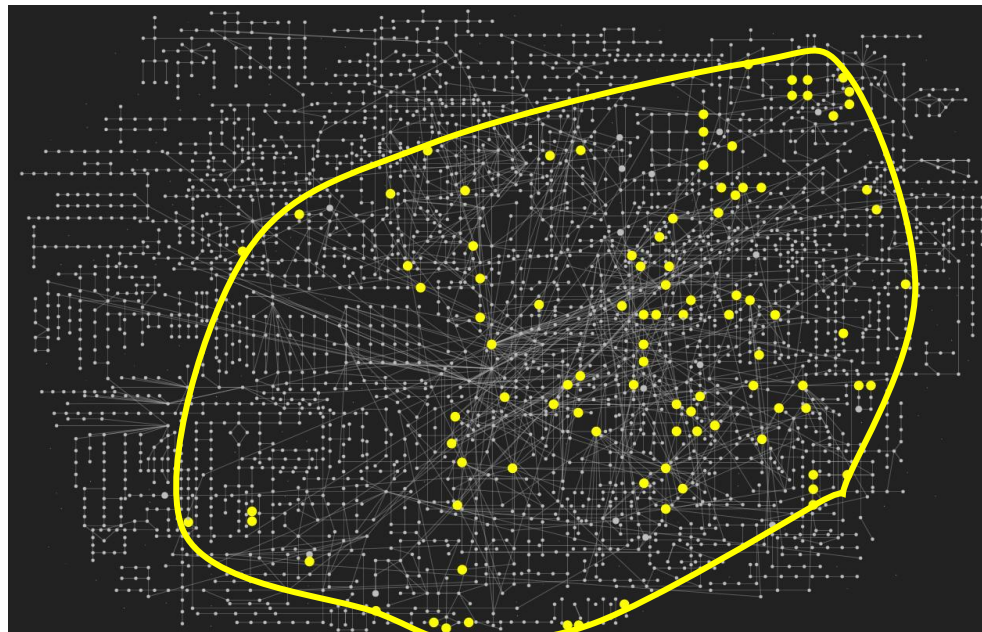
Biosynthesis of Secondary Metabolites

Individual Metabolites overlap between BIOCRATES and BIOMET (inchikey ID comparison)

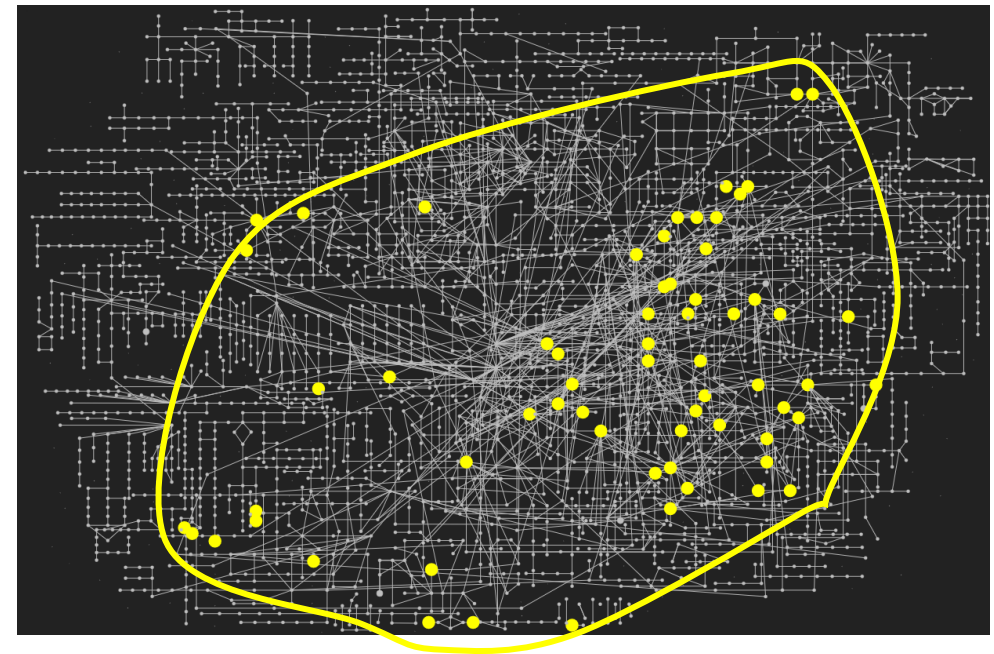


Pathway coverage comparison between BIOMET and BIOCRATES analyses

BIOMET

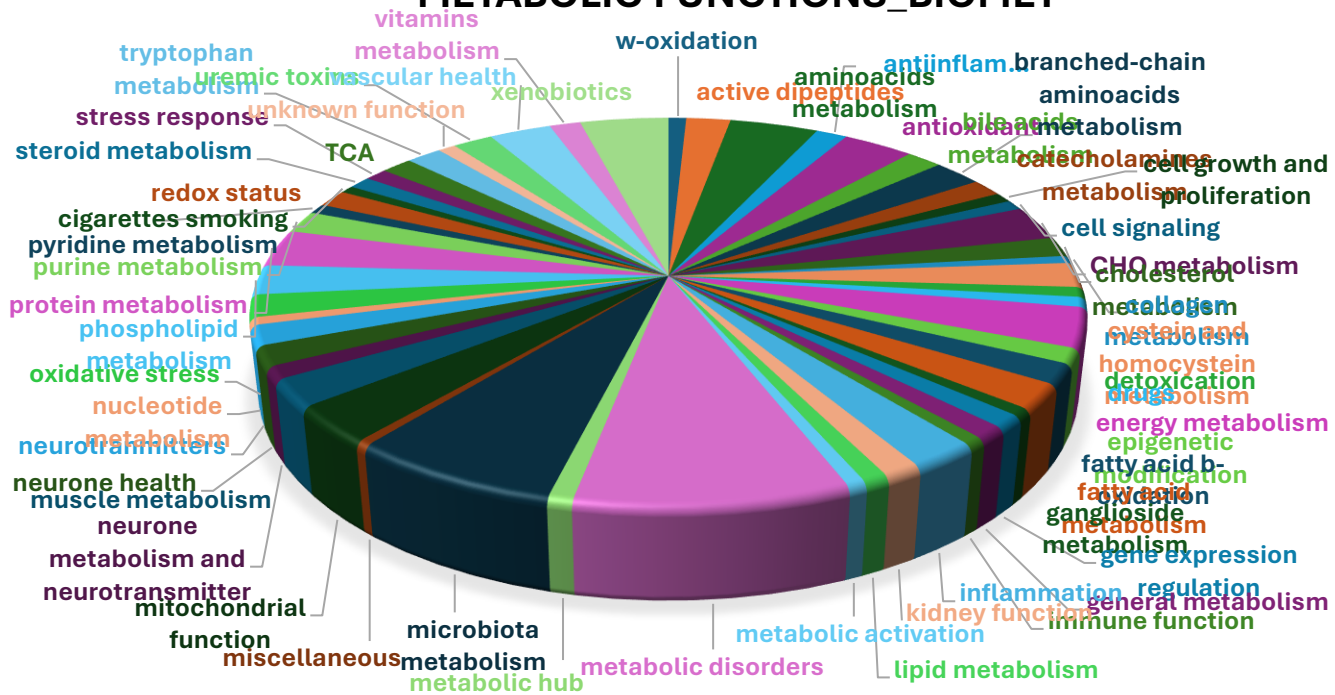


BIOCRATES

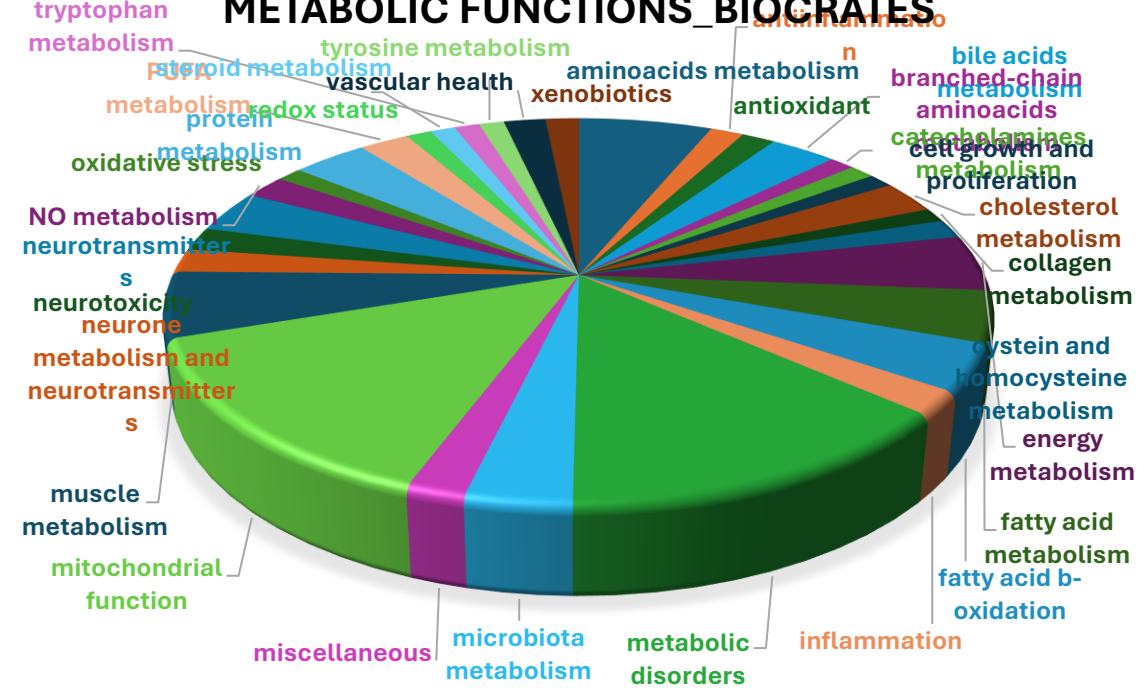


METABOLOMICS ANALYSES BY BIOLOGICAL FUNCTIONS

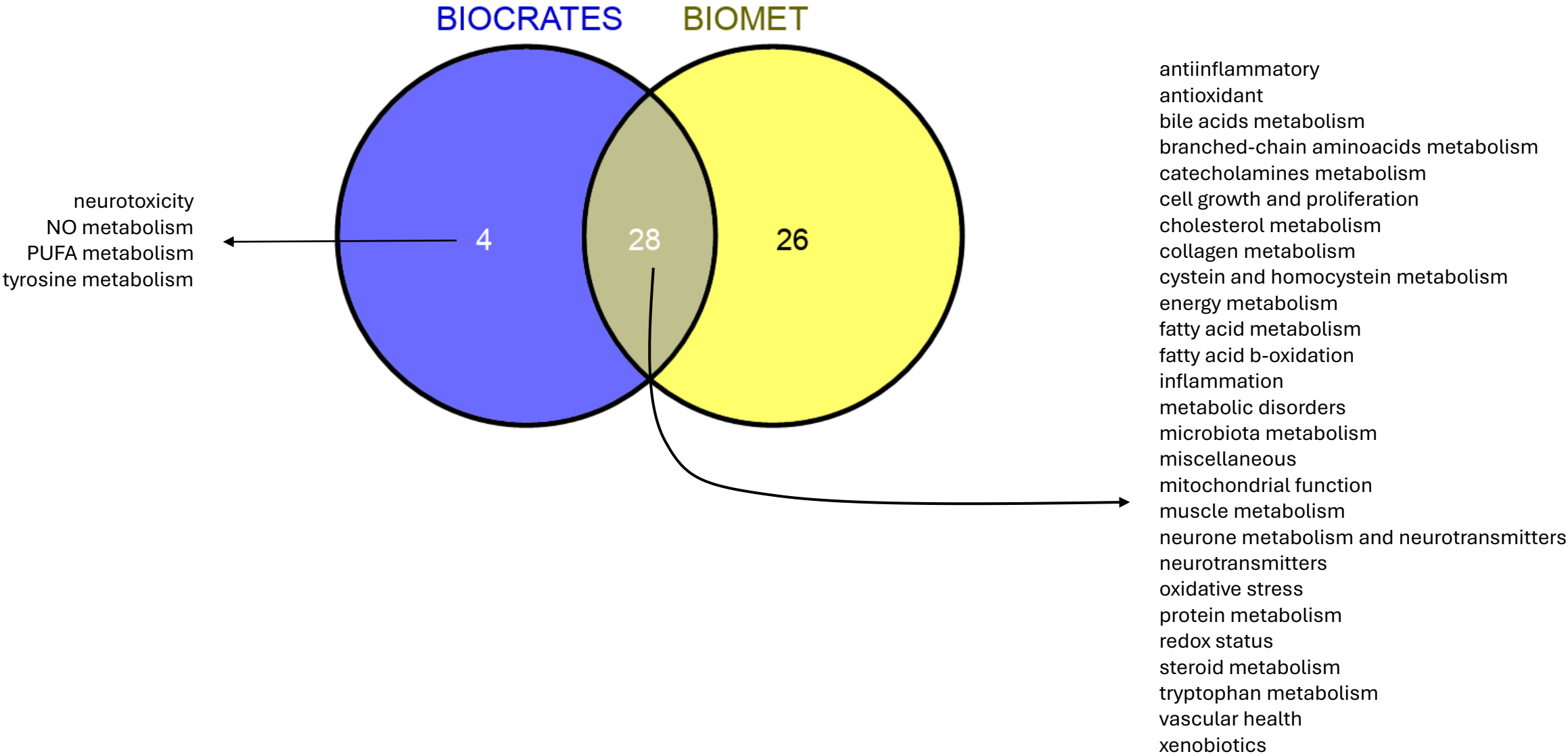
METABOLIC FUNCTIONS_BIOMET



METABOLIC FUNCTIONS_BIOCRATES

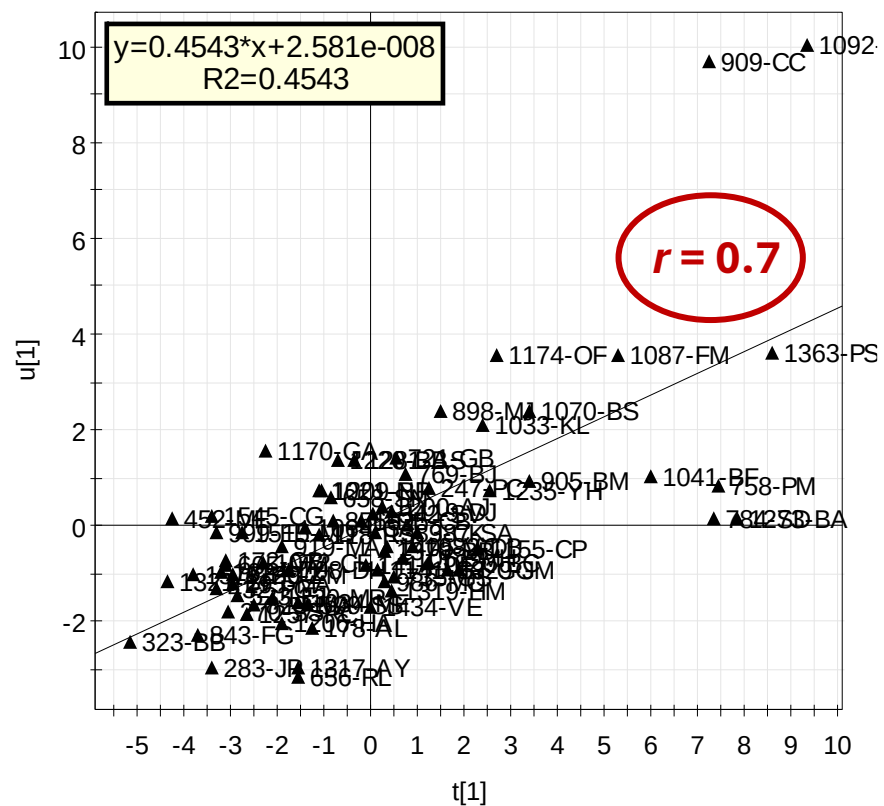


Metabolic functions overlaps between BIOCRATES and BIOMET



Correlations between shared biological functions between biocrates and biomet

merged MRS and biocrates lipido_metabo blocks.M6 (OPLS/O2PLS), shared biological functions biomet_biorcrates (UV scale)
t[Comp. 1]/u[Comp. 1]





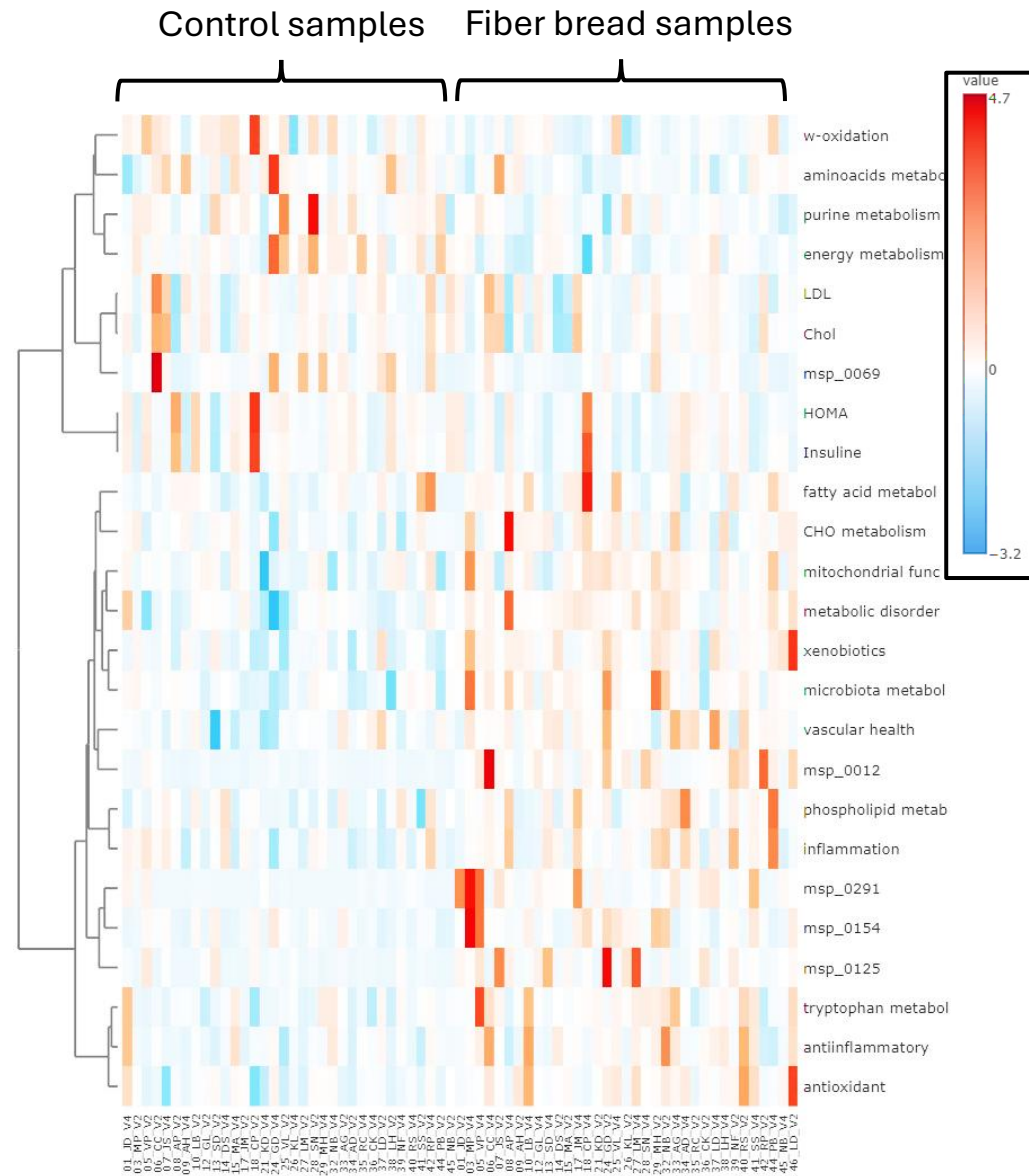
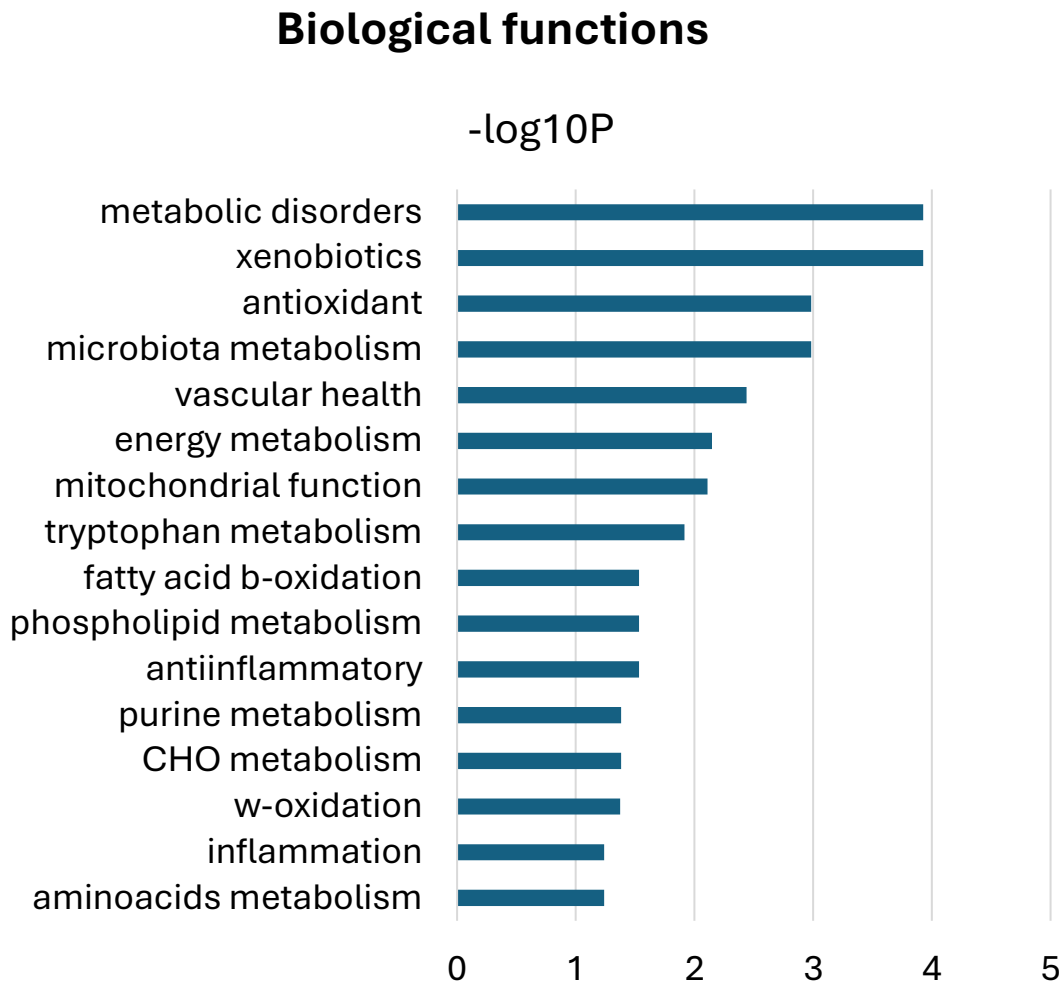
Identification of the biological response induced by the consumption of a multi-fiber bread using a deep phenotyping approach in human volunteers with the metabolic syndrome

Partnership JA Nazarre, CARMEN & Metagenopolis

Deep phenotyping = Metabolomics x metagenomics x clinical variables

Statistically impacted biological functions in multi-fiber consumers

Sorting according to biological impact



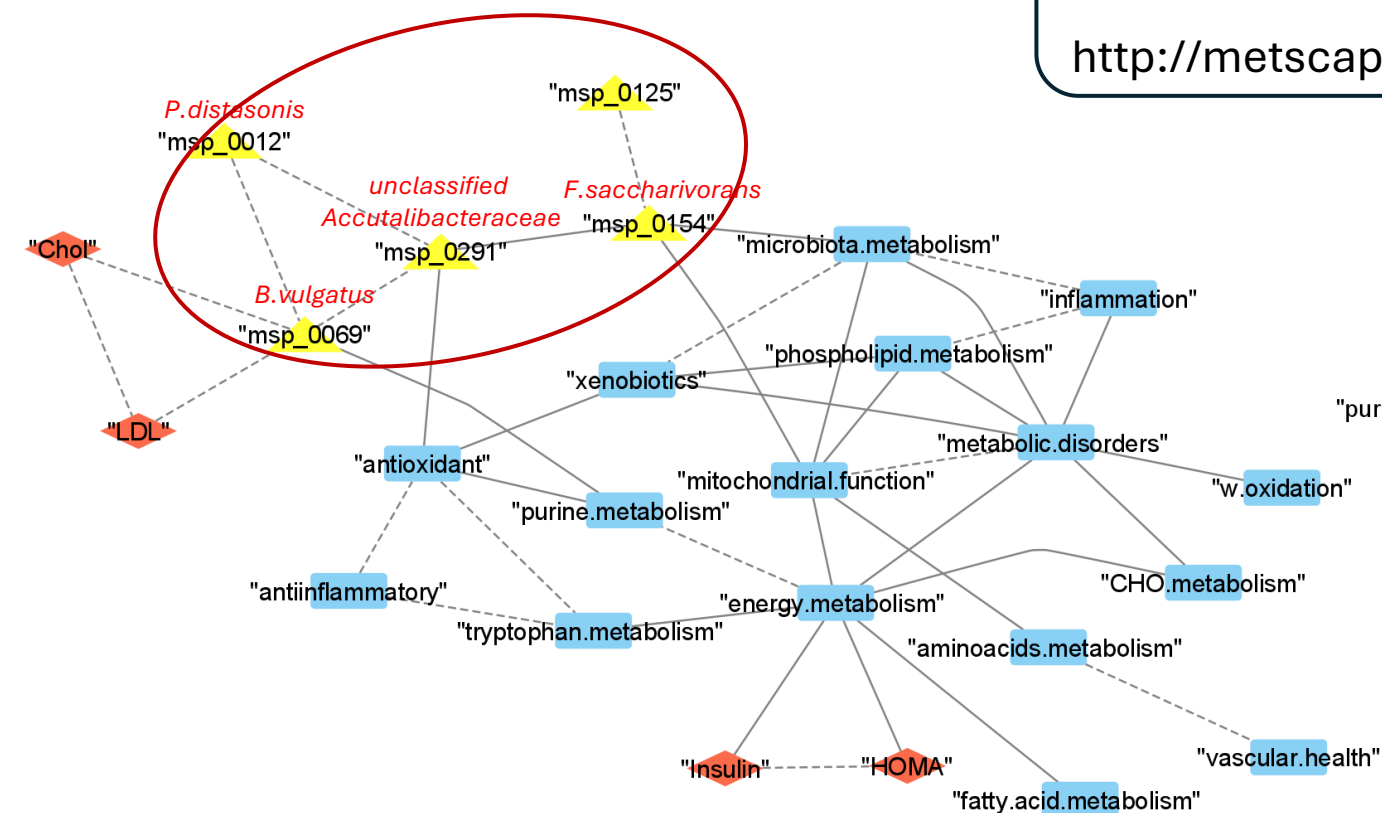
Multilevel analysis => integrating clinics, metagenomics, metabolomics



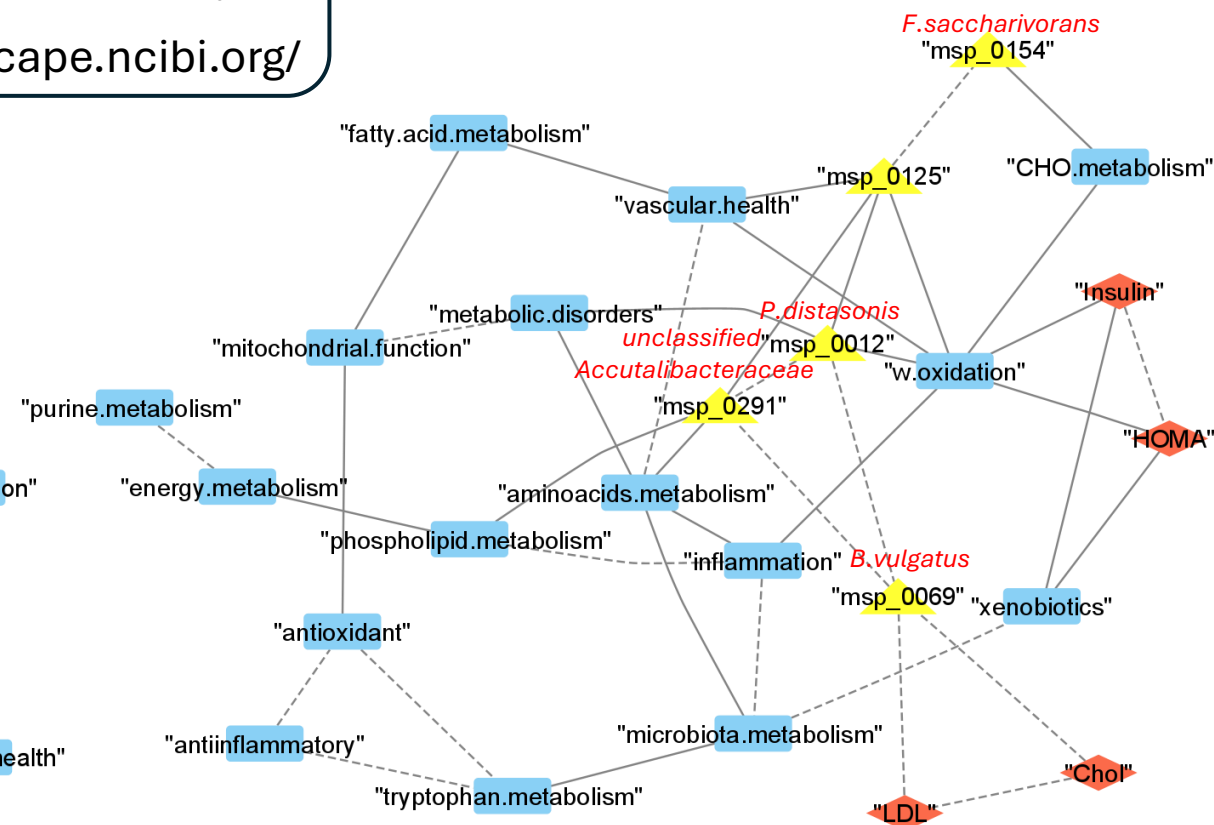
CorrelationCalculator-1.0.1.jar

<http://metscape.ncibi.org/>

Fiber bread consumers network



Control individuals network



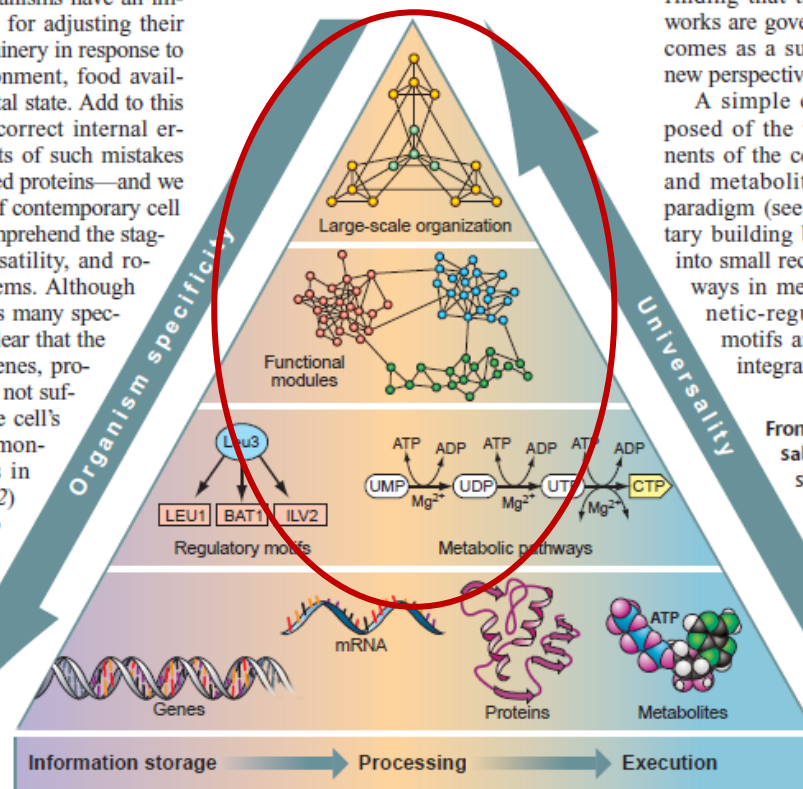
Life's Complexity Pyramid

Zoltán N. Oltvai and Albert-László Barabási

Cells and microorganisms have an impressive capacity for adjusting their intracellular machinery in response to changes in their environment, food availability, and developmental state. Add to this an amazing ability to correct internal errors—battling the effects of such mistakes as mutations or misfolded proteins—and we arrive at a major issue of contemporary cell biology: our need to comprehend the staggering complexity, versatility, and robustness of living systems. Although molecular biology offers many spectacular successes, it is clear that the detailed inventory of genes, proteins, and metabolites is not sufficient to understand the cell's complexity (1). As demonstrated by two papers in this issue—Lee *et al.* (2) on page 799 and Milo *et al.* (3) on page 824—viewing the cell as a network of genes and proteins offers a viable strategy for addressing the complexity of living systems.

According to the basic dogma of molecular biology, DNA is the ultimate depository of biological complexity. Indeed, it is generally accepted that information storage, information processing, and the execution of various cellular programs reside in distinct levels of organization: the cell's genome, transcriptome, proteome, and

Z. N. Oltvai is in the Department of Pathology, Northwestern University, Chicago, IL 60611, USA. E-mail: zno008@nwu.edu A.-L. Barabási is in the Department of Physics, University of Notre Dame, Notre Dame, IN 46556, USA. E-mail: alb@nd.edu



metabolome. However, the distinctness of these organizational levels has recently come under fire. For example, although long-term information is stored almost exclusively in the genome, the proteome is crucial for short-term information storage (4, 5) and transcription factor-controlled information retrieval is strongly influenced by the state of the metabolome. This integration of different organizational levels increasingly forces us to view cellular functions as distributed among groups of heterogeneous components that all interact

within large networks (6, 7). There is clear evidence for the existence of such cellular networks: For example, the proteome organizes itself into a protein interaction network and metabolites are interconverted through an intricate metabolic web (7). The finding that the structures of these networks are governed by the same principles comes as a surprise, however, offering a new perspective on cellular organization.

A simple complexity pyramid composed of the various molecular components of the cell—genes, RNAs, proteins, and metabolites—summarizes this new paradigm (see the figure). These elementary building blocks organize themselves into small recurrent patterns, called pathways in metabolism and motifs in genetic-regulatory networks. In turn, motifs and pathways are seamlessly integrated to form functional mod-

From the particular to the universal. The bottom of the pyramid shows the traditional representation of the cell's functional organization: genome, transcriptome, proteome, and metabolome (level 1). There is remarkable integration of the various layers both at the regulatory and the structural level. Insights into the logic of cellular organization can be achieved when we view

the cell as a complex network in which the components are connected by functional links. At the lowest level, these components form genetic-regulatory motifs or metabolic pathways (level 2), which in turn are the building blocks of functional modules (level 3). These modules are nested, generating a scale-free hierarchical architecture (level 4). Although the individual components are unique to a given organism, the topologic properties of cellular networks share surprising similarities with those of natural and social networks. This suggests that universal organizing principles apply to all networks, from the cell to the World Wide Web.

gation
d
hesis

vascular outcomes

s

analamine
sis

vascular health
Xenobiotics
processing

xenobiotics

Caffeine
metabolism

Caffeine and
Theobromine metabolism

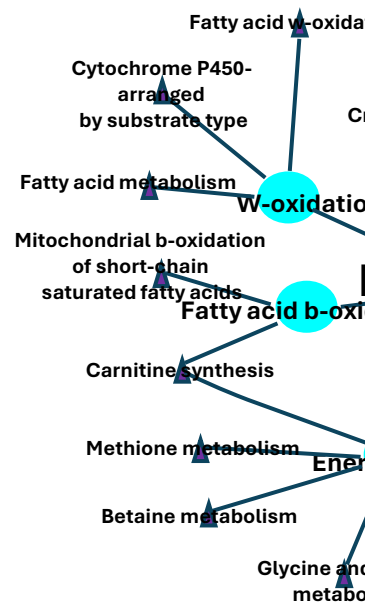
inflammatory

n

Biological system

Biological outcomes

Biochemical pathways



Niveau moléculaire (information biochimique)

Niveau fonctionnel (information processus biologiques)

Niveau systèmes (Information sur la réponse phénotypique)

Tryptophan metabolism
Gut-liver indole metabolism
Farnesyl to CoQ10 metabolism
Amine Oxidase reactions

Phosphate co-transporter
Sodium-coupled phosphate cotransporters

Thiamine metabolism
Creatine pathway
SLC-mediated transmembrane transport
Ammonia recycling

Methylhistidine metabolism
Betaine metabolism
Methionine metabolism

SLC-mediated transmembrane transport
Purine metabolism
Nucleotide salvage
Transport of small molecules

Vitamin B6 metabolism
Steroid hormone biosynthesis
Histidine metabolism
Steroidogenesis

2 (2-AMINOETHYL)PHOSPHONATE
2 3-INDOLEPROPIONIC Acid
3-(2,4-Dihydroxyphenyl)propionic acid
3-(3-HYDROXYPHENYL)PROPIONIC acid
3,4-DIHYDROXYHYDROCINNAMIC Acid
3-AMINO-4-HYDROXYBENZOATE

Phosphoric acid
TRIGONELLINE
Mevalonolactone

OROTATE
PANTOTHENATE
PHOSPHOCREATINE
Phosphoric acid
PYRIDOXINE
RIBOFLAVIN

GLYCINE
Histamine
L-ORNITHINE
HYPOXANTHINE
INOSINE
Phosphoric acid
CYTIDINE
1-METHYLADENOSINE
3-PHENYLLACTIC Acid
FUCOSE
Histamine
INDOLE 3-CARBOXALDEHYDE
LYSINE
MANNOSE
PYRIDOXAL

**Microbiota
metabolism**

kidney function

MetaboSmart™
Metabolomics Made Smart

epigenetic modification

hemostasis

immune function

Microbiota metabolic systems

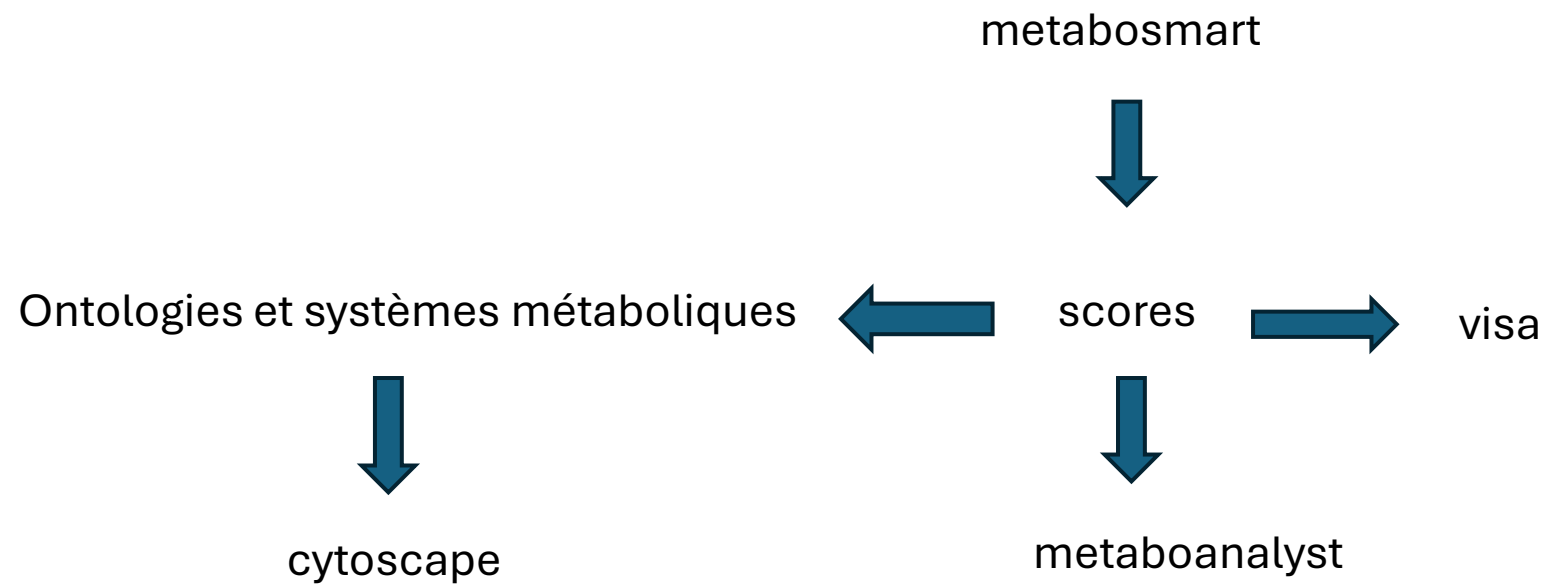
Organs and tissues systems

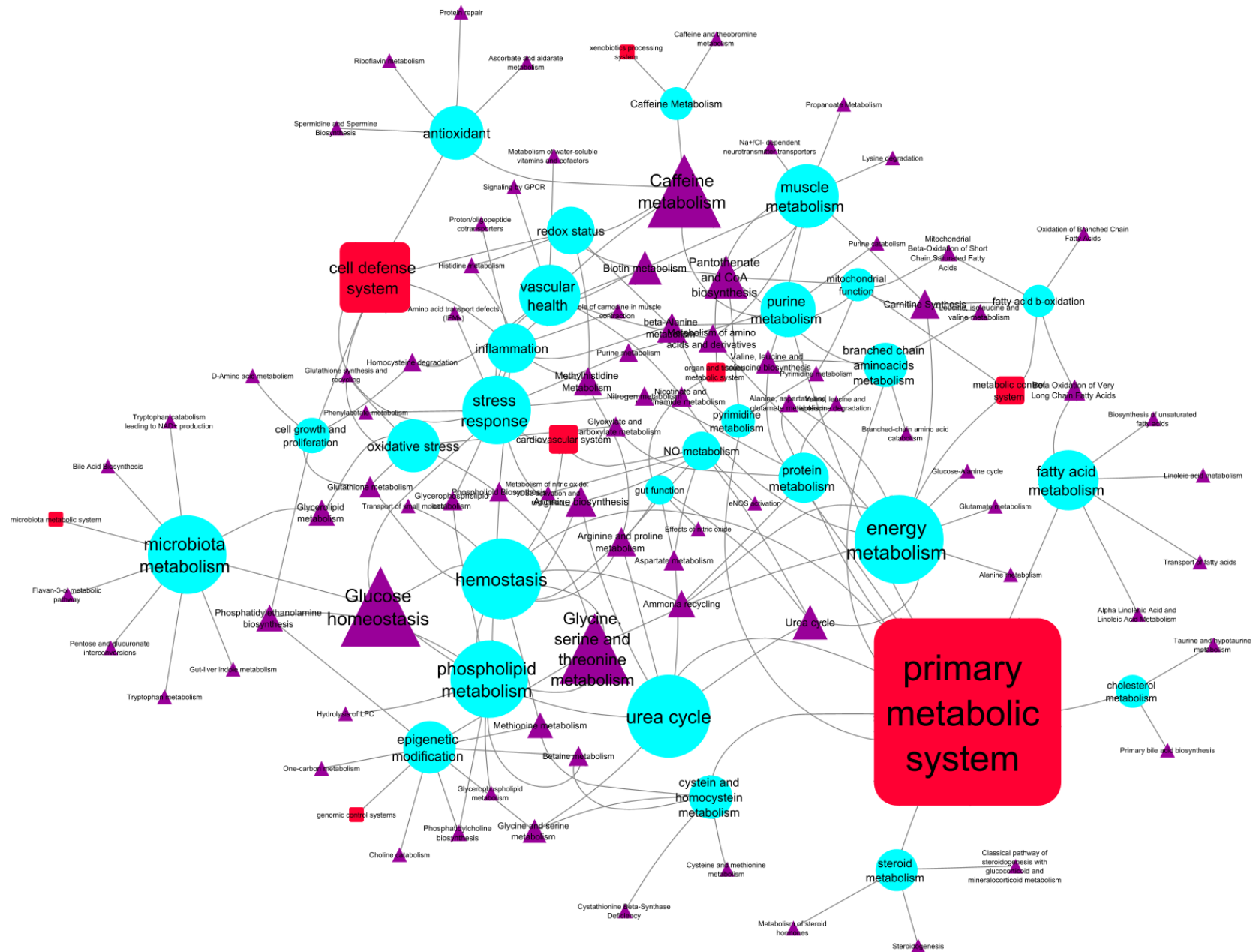
metabolic control systems

Genomics systems

Cardiovascular systems

Cell defense systems

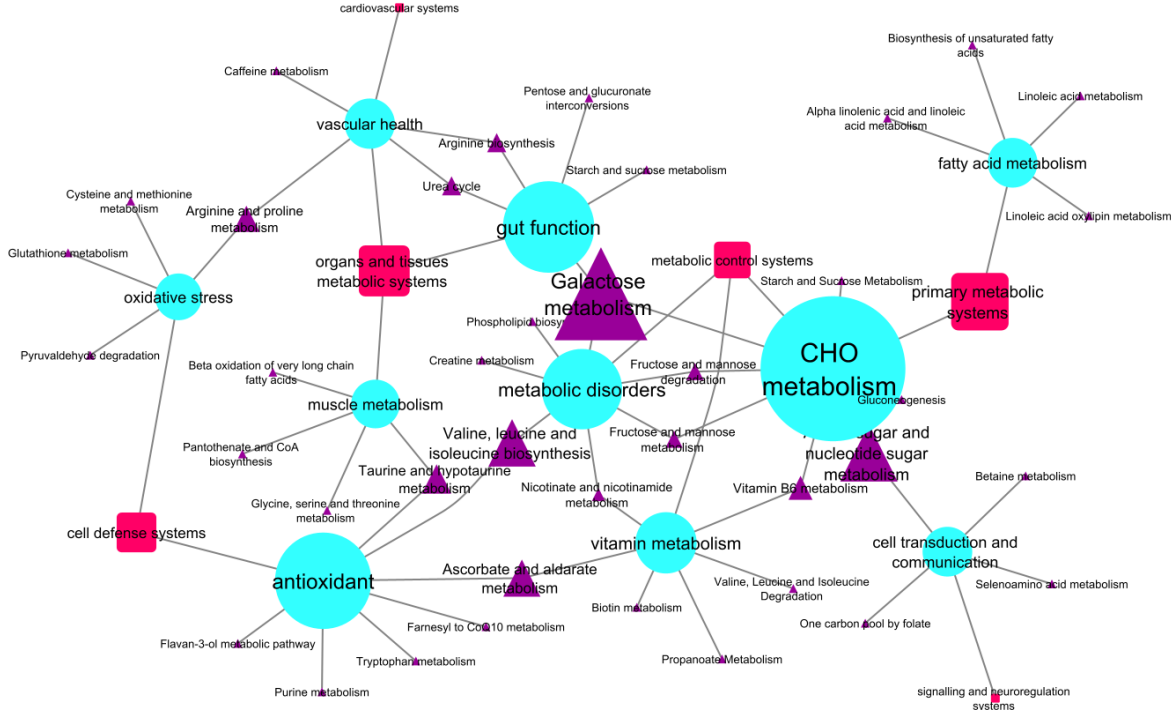
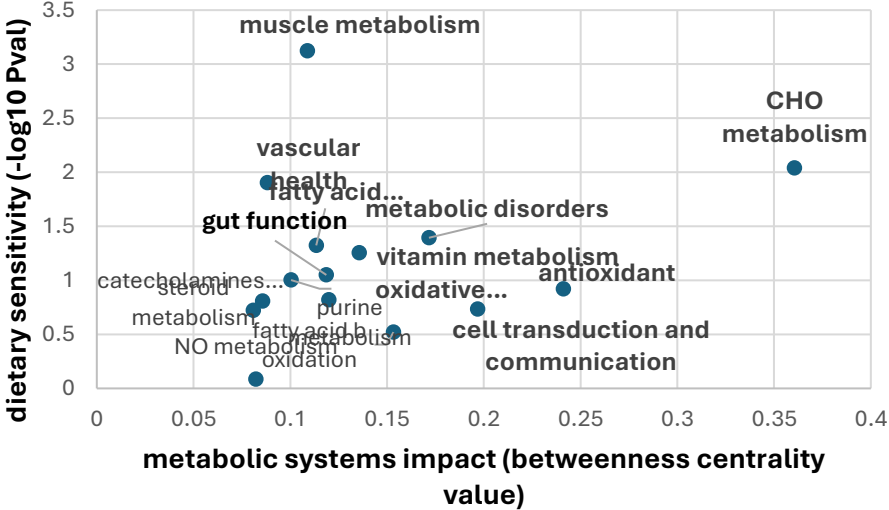




Individual Spearman rank correlation *P*-value to IsoP score

oxidative stress	0.001
fatty acid b oxidation	0.002
antioxidant	0.002
catecholamines metabolism	0.005
vascular health	0.007
CHO metabolism	0.008
fatty acid metabolism	0.009
metabolic disorders	0.016
cell transduction and communication	0.019
w oxidation	0.026
vitamin metabolism	0.033
purine metabolism	0.042
muscle metabolism	0.044
gut function	0.052
NO metabolism	0.061
steroid metabolism	0.087

Functions impact





MetaboSmartTM
Metabolomics Made Smart

REF DATABASE
(split metabolites into functions)



Automated AI generated (LLM search) (plant use,
microorganisms, custom)

Includes mining into pathways databases



Automated ontology network analysis